M3: Complete Implementation

CS 4412 — Data Mining

1 Overview: Where You Are in the Project

In M2, you explored your data and applied your first mining technique. Now in M3, you bring your implementation to **functional completeness**—your analysis pipeline works end-to-end, you've expanded beyond M2, and you have substantial results to show.

By the end of M3, you should be able to say: "My code runs. My analysis produces meaningful results. I've explored my options and made informed choices about my approach. The technical work is essentially done—what remains is polish and presentation."

What "Complete Implementation" Means

Complete means your implementation is functionally done:

- All data pipelines working (ingestion, cleaning, transformation)
- All techniques you're going to use are implemented and running
- Discovery questions are stable—not changing scope anymore
- You have results, even if they need refinement

Not yet complete (that's M4):

- Final polish and presentation-ready formatting
- Critical assessment of validity and limitations
- Portfolio-quality documentation

1.1 What You've Covered in Class

By M3's due date, you've been introduced to:

- Everything from M2 (EDA, preprocessing, association rules, clustering)
- Classification: Decision Trees
- Classification: Naive Bayes
- Dimensionality Reduction: PCA
- Anomaly Detection

M3 asks you to demonstrate **implementation completeness** and **meaningful analytical expansion** beyond M2.

2 Analytical Expansion

M3 requires meaningful growth beyond M2. This doesn't mean checking boxes—it means demonstrating that you've explored your analytical options and made informed choices.

2.1 What Counts as Expansion

Your analysis should demonstrate meaningful growth from M2. This could take several forms:

- **New technique category**: Adding classification to your clustering work, or anomaly detection to your association rules
- **Multiple approaches within a category**: Comparing K-Means, DBSCAN, and hierarchical clustering to understand which fits your data best
- **Deeper exploration of M2 technique**: Meaningful variations—different parameters, feature subsets, applied to different segments of your data
- **Technique appropriate to your data**: Something we haven't explicitly covered but that fits your discovery questions (e.g., sequence mining, graph analysis)

Guidance

The goal is evidence that you've *explored* and made *informed choices*—not a checklist of algorithms. Someone who tries three clustering approaches, compares them thoughtfully, and explains why DBSCAN fits their data best has demonstrated more analytical maturity than someone who runs K-Means and Decision Trees without understanding why.

2.2 Technique Options

Here are techniques available to you by M3:

Category	Techniques
Association Rules	Apriori, FP-Growth
Clustering	K-Means, Hierarchical, DBSCAN
Classification	Decision Trees, Naive Bayes
Anomaly Detection	Distance-based, Density-based (LOF)
Dimensionality Reduction	PCA (for visualization or as preprocessing)

This list is not exhaustive. If a different technique is appropriate for your data and discovery questions, use it.

2.3 Choosing Your Expansion

Let your discovery questions guide technique selection:

If you're asking...	**Consider adding...**
"What types/segments exist?"	Clustering (if not done), or Classification to characterize segments
"What features distinguish groups?"	Decision Trees (interpretable rules), PCA (visualize separation)
"What co-occurs or associates?"	Association Rules, or Clustering on item vectors
"What's unusual or anomalous?"	Anomaly Detection (LOF, distance-based)
"How can I visualize high-dimensional patterns?"	PCA for dimensionality reduction

3 Classification in a Discovery Context

Classification is typically framed as prediction: given features, predict a class label. In this course, we use classification differently—for **pattern discovery through interpretable rules**.

3.1 Decision Trees for Discovery

Decision trees produce human-readable rules. The value isn't prediction accuracy—it's the *rules themselves*:

- What features does the tree split on first? (These are most informative.)
- What rules characterize each class?
- Are the rules surprising or expected?

Guidance

A shallow decision tree (depth 3–5) often produces more interpretable rules than a deep one. You're not trying to maximize accuracy—you're trying to understand what distinguishes classes.

3.2 Naive Bayes for Discovery

Naive Bayes estimates P(class | features). For discovery:

- Which features have the strongest association with each class?
- What does the conditional probability structure reveal?
- How do the class priors compare to what you expected?

3.3 Where Do Class Labels Come From?

If your data doesn't have natural class labels, you can:

- **Use cluster assignments from M2** as class labels—then build a decision tree to explain what distinguishes the clusters
- **Discretize a target variable**—e.g., "high" vs. "low" sales

- **Use an existing categorical variable**—e.g., product category, region, customer type

Common Pitfall

If you use cluster labels as classes, you're not doing prediction—you're using classification as an *interpretability tool* to explain your clusters. This is legitimate and often valuable. Just be clear about what you're doing.

4 Anomaly Detection

Anomaly detection identifies unusual observations. In a discovery context, anomalies can be:

- **Errors** worth investigating or removing
- **Interesting edge cases** that reveal something about your domain
- **Potential fraud, outliers, or rare events** depending on your data

4.1 Approaches

- **Distance-based**: Points far from their neighbors are anomalies. Simple to implement, but sensitive to parameter choices.
- **Density-based (LOF)**: Points in low-density regions relative to their neighbors are anomalies. More robust to varying cluster densities.

4.2 Interpretation

Don't just flag anomalies—investigate them:

- What makes these points unusual?
- Are they data errors, or genuinely interesting?
- Do they cluster together (suggesting a hidden subgroup)?
- What do they reveal about the boundaries of "normal"?

5 Dimensionality Reduction (PCA)

PCA can serve two purposes in your analysis:

5.1 Visualization

If you have many features, PCA can reduce to 2–3 dimensions for plotting. This helps you:

- Visualize cluster structure
- See class separation
- Identify outliers visually

5.2 Preprocessing

PCA can reduce dimensionality before applying other techniques:

- Reduces noise and multicollinearity
- Can improve clustering or classification on high-dimensional data
- Makes subsequent analysis faster

Guidance

When using PCA, report how much variance is captured by your chosen components. "The first two principal components capture 75% of the variance" tells the reader how much information is preserved.

6 Exploration and Informed Choices

If you've tried multiple approaches, comparing them helps you understand your data and justify your choices.

6.1 Questions to Guide Comparison

- **Consistency**: Do different techniques find similar patterns? If so, your findings are more robust.
- **Complementarity**: Do different techniques reveal different aspects? Both might be valuable.
- **Appropriateness**: Which technique best addresses each discovery question? Why?
- **Trade-offs**: What does each technique assume? Where does it struggle with your data?

Common Pitfall

Comparison isn't about declaring a "winner." K-Means isn't "better" than DBSCAN—they answer different questions and make different assumptions. Compare what each *reveals*, not which "wins."

6.2 Documenting Your Choices

In your analysis document, explain:

- What approaches did you try?
- What did you learn from each?
- Why did you settle on your final approach?
- What alternatives did you consider and reject (and why)?

This kind of reasoning—showing you understand the landscape of options, not just the one algorithm you ran—is what distinguishes strong analytical work.

7 Pattern Discovery and Results

M3 is where your results become substantial. You should have actual findings—patterns, clusters, rules, anomalies—not just code that runs.

7.1 What Counts as Substantial Results

- **Association Rules**: A set of interesting rules with support, confidence, and lift. Not just "I ran Apriori"—what rules did you find? Which are meaningful vs. trivial?
- **Clustering**: Identified clusters with characterization. Not just "K-Means found 4 clusters"—what defines each cluster? Can you name them?
- **Classification**: Interpretable rules or feature importances. What does the decision tree reveal about what distinguishes classes?
- **Anomaly Detection**: Identified anomalies with analysis. What makes them unusual? Are they errors or interesting edge cases?

7.2 Connecting to Discovery Questions

Revisit your M1 proposal. For each discovery question:

- What have you learned so far?
- Which techniques addressed this question?
- What's the answer (even if partial)?
- What remains unclear?

7.3 Interpretation

Translate statistical findings into domain language:

- Not: "Cluster 2 has mean = 4.7 for feature X"
- But: "Cluster 2 represents price-sensitive customers who purchase frequently but in small amounts"

Looking Ahead to M4

M3 is your working implementation. M4 is your polished final product. Use M3 to identify what's working, what needs refinement, and what story you want to tell. The technical heavy lifting should be done; M4 is about polish, critical assessment, and presentation.

8 Deliverables

Submit the following to D2L with your GitHub repository link in the comments:

8.1 Complete Analysis Notebook(s)

Your notebook(s) should demonstrate a working end-to-end implementation:

1. Data loading, cleaning, and preprocessing (updated from M2 if needed)
2. All mining techniques you're using—implemented and producing results
3. Visualizations of your findings
4. Interpretation connecting results to discovery questions

The notebook should be **runnable**—code executes, outputs display. Someone should be able to clone your repo and reproduce your results.

8.2 Analysis Document (PDF)

A 2–4 page document containing:

1. **Implementation Status**: What's working? What techniques did you apply?
2. **Analytical Choices**: What approaches did you explore? Why did you choose your current approach?
3. **Results**: What patterns did you discover? What do they mean?
4. **Discovery Question Progress**: How are you answering your M1 questions?
5. **M4 Plan**: What remains for polish and presentation?

8.3 GitHub Repository

Your repository should include:

- All code and notebooks (organized clearly)
- Data files or access instructions
- Updated README reflecting M3 progress
- Clear folder structure

9 References and Resources

9.1 Textbook

Zaki & Meira, *Data Mining and Machine Learning* (free PDF at https://dataminingbook.info):

- **Chapter 7**: Dimensionality Reduction (pp. 183–214) — PCA
- **Chapter 18**: Probabilistic Classification (pp. 469–490) — Naive Bayes
- **Chapter 19**: Decision Tree Classifier (pp. 491–500)
- **Chapter 22**: Classification Assessment (pp. 549–582) — evaluation metrics

For anomaly detection, see Aggarwal, *Outlier Analysis* (recommended text):

- Chapter 3: Distance-based methods
- Chapter 4: Density-based methods (LOF)

9.2 Python Libraries

- `scikit-learn`: `DecisionTreeClassifier`, `GaussianNB`, `PCA`, `LocalOutlierFactor`
- `pyod`: Comprehensive anomaly detection library (https://pyod.readthedocs.io)

9.3 Tutorials

- Scikit-learn Decision Trees: https://scikit-learn.org/stable/modules/tree.html
- Scikit-learn PCA: https://scikit-learn.org/stable/modules/decomposition.html#pca
- PyOD documentation: https://pyod.readthedocs.io/en/latest/

10 Grading

See D2L for the detailed grading rubric. Key evaluation areas:

- **Implementation Completeness**: Does your analysis pipeline work end-to-end?
- **Analytical Expansion**: Have you grown meaningfully beyond M2?
- **Pattern Discovery**: Do you have substantial results—actual findings, not just code?
- **Discovery Question Progress**: Are you answering your M1 questions?
- **Documentation**: Is your analysis document clear? Is your code organized?

The focus is on functional completeness and evidence of informed analytical choices. M4 will assess polish, critical assessment, and presentation readiness.

Questions? Attend office hours or email following the course communication policy.